

Vision

The ability to see allows all animals to build up a detailed picture of their surroundings—vital for finding food and avoiding danger.

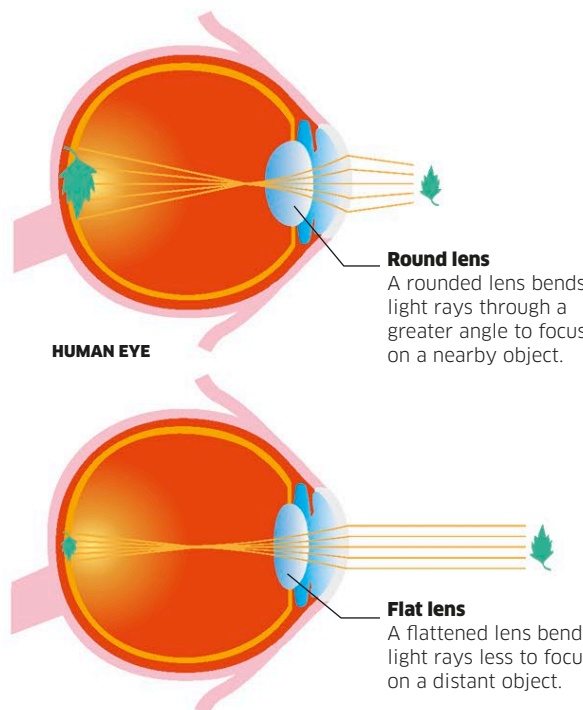
When an animal sees the world, its eyes pick up light and use lenses to focus this onto light-sensitive receptor cells. These cells then send signals to the brain, which composes a visual image of everything in the field of view. For animals with the best vision, the image can be finely detailed—even when the light is poor.

Night eyes

The eyes of birds are so big in proportion to their head that they are largely fixed inside their sockets. This means a bird must rotate its flexible neck to look around. Owl eyes, like those of many nocturnal birds, are especially large and are designed for good night vision. Their unusual shape creates room for a larger space at the back of the eye, packed with extra light-sensitive cells.

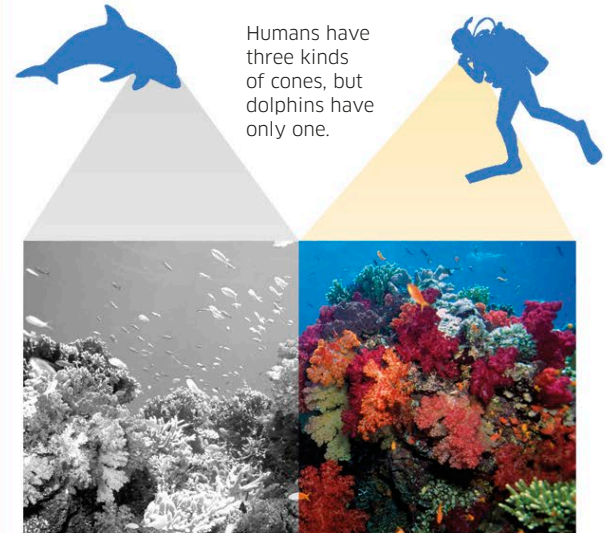
Near and far

The eye's lens focuses light onto the retina, and can change shape to better focus on either closer objects or those farther away. A ring of muscle controls this shape. It contracts to make the lens rounder for near focus, and relaxes to pull the lens flatter for distant focus.



Seeing color

Receptor cells called cones are what allow animals to see color. These detect different light wavelengths—from short blue wavelengths to long red ones. Animals with more types of cones can see more colors, but those with just one are only able to see the world in black and white.



Three cones help humans see three primary colors: red, green, and blue, plus all their combinations.

Many day-flying birds have one more type of cone than humans, meaning they can see ultraviolet.

Binocular vision

When two forward-facing eyes have overlapping fields of view, this is called binocular vision. This gives an animal a three-dimensional view of the world, helping it to judge distance—a skill especially important for predators that hunt prey. Other animals with eyes on the sides of their head have a narrower range of binocular vision, but better all-around vision.

